

GC carbon calibration for SWING data

$$I_{GC} (cm^{-1}) = K \frac{I_{Swing_GC} - I_{Swing_air}}{l_{GC} (mm)}$$

In Foxtrot macro, use
K=0.036 for the calibration
constant (instead of K=1)

GC calibration: K=0.036

Superimpose GC (cm-1) with the calibration curve by Jan Ilavsky (file: GlassyCarbon_T.dat).
Calibration constant K should be close to 0.036.

$l_{GC} = 1.16$ mm

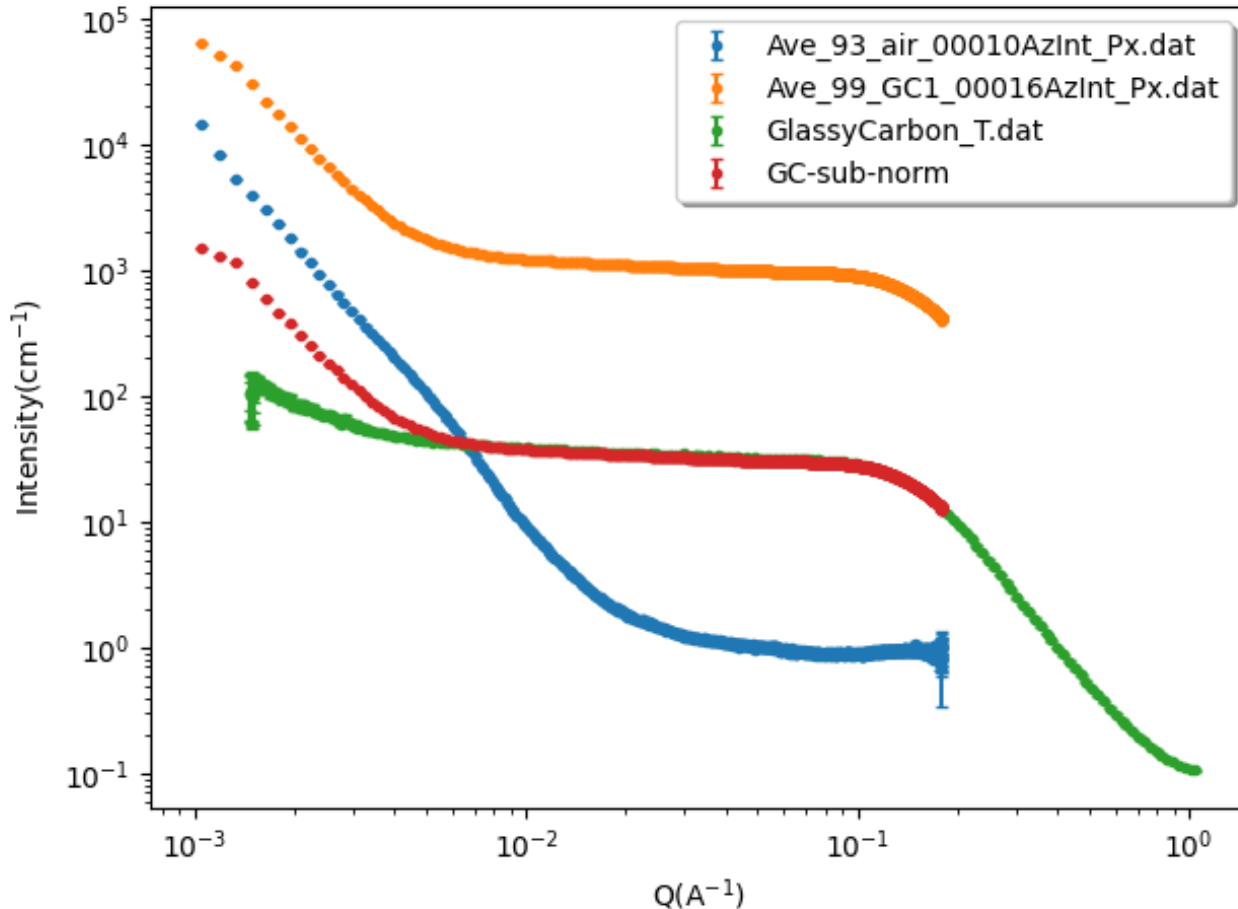
$$I_{cap_norm_sub} (cm^{-1}) = K \frac{I_{Swing_cap} - I_{Swing_cap_solvent}}{l_{cap} (mm)}$$

Calibration of intensity for a sample in a glass capillary:

Take the same thickness l_{cap} (in mm) for the capillary filled with solvent

Thickness is not taken into
account in Foxtrot data
reduction

SWING data April 26: GC calibration



$l_{GC} = 1.16 \text{ mm}$

GC calibration: $K=0.036$

$$I_{GC} (\text{cm}^{-1}) = K \frac{I_{Swing_{GC}} - I_{Swing_{air}}}{l_{GC} (\text{mm})}$$

In SASView GUI, go to tools/data operation:

- Subtract air signal:
-> Compute and save $I_{Swing_{GC}} - I_{Swing_{air}}$
- Divide by thickness $l_{GC} = 1.16 \text{ mm}$:
-> Compute and save $\frac{I_{Swing_{GC}} - I_{Swing_{air}}}{l_{GC} (\text{mm})}$
- Multiply by calibration constant **$K=0.036$**
-> Compute and save $I_{GC} (\text{cm}^{-1}) = K \frac{I_{Swing_{GC}} - I_{Swing_{air}}}{l_{GC} (\text{mm})}$
- Verify that $I_{GC} (\text{cm}^{-1})$ superimposes with the calibration curve by Jan Ilavsky (file: GlassyCarbon_T.dat)

For all SWING data

In Foxtrot, use K=0.036 for the calibration constant (instead of K=1) and run again the macro

$$I_{cap_norm_sub} (cm^{-1}) = K \frac{I_{Swing_cap} - I_{Swing_cap_solvent}}{l_{cap} (mm)} = K \frac{I_{Swing_cap}}{l_{cap} (mm)} - K \frac{I_{Swing_cap_solvent}}{l_{cap} (mm)}$$

$$I_{cap_norm} (cm^{-1}) = K \frac{I_{Swing_cap}}{l_{cap} (mm)}$$

Calibration of intensity for a sample in a glass capillary:

** Take the same thickness l_{cap} (in mm) for the capillary filled with solvent